

UI/UX Screen Specification — Examination & Result Management Module

Reuses RGD v2.0, Appendix A-N, and the approved Admission, SIS, and Attendance UI/UX Screen Specifications by reference only; none regenerated. FR-17 to FR-21, BR-EXM-001 to BR-EXM-007, and the Exam/MarksRecord/ReportCard/GradingScheme entities (Appendix-G/H) are cited by ID.

#	Screen ID	Screen Name
1	SCR-EXM-001	Exam Dashboard
2	SCR-EXM-002	Exam Master
3	SCR-EXM-003	Exam Schedule
4	SCR-EXM-004	Subject Marks Entry
5	SCR-EXM-005	Bulk Marks Upload
6	SCR-EXM-006	Marks Verification
7	SCR-EXM-007	Result Processing
8	SCR-EXM-008	Grade Configuration
9	SCR-EXM-009	Rank Calculation
10	SCR-EXM-010	Report Card Generation
11	SCR-EXM-011	Publish Results
12	SCR-EXM-012	Result Search
13	SCR-EXM-013	Student Result History
14	SCR-EXM-014	Result Analytics

SCR-EXM-001 — Exam Dashboard

Purpose	Landing summary of upcoming exams, pending marks entry, and pending verification.
User Roles	Academic Head, Teacher, Admin Staff, IT Admin, Management (read-only)
Navigation Path	Examination > Dashboard
Page Header	"Examination Dashboard"
Breadcrumb	Home > Examination > Dashboard
Form Sections	Upcoming Exams, Pending Marks Entry, Pending Verification, Quick Links
Field Display Order	Upcoming Exams (widget) -> Pending Marks Entry (widget) -> Pending Verification (widget) -> Quick Links
Conditional Visibility Rules	"Pending Verification" widget shown only to Academic Head.
UI Behaviour	Widgets refresh on load; no inline editing.
Buttons & Actions	Go to Marks Entry, Go to Verification, View Reports
Button Placement	Widget-level action links.
Toolbar Actions	N/A
Table Columns	N/A (widget-based)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	N/A
Filters	Academic Session
Default Sorting	N/A
Default Pagination	N/A
Bulk Actions	N/A
Export Formats	N/A
Print Behaviour	N/A
Empty State	"No exams scheduled for this session."
Loading State	Skeleton widgets while data loads.
Popup/Dialog Details	N/A
Audit Information	N/A (summary view).

Responsive Behaviour	Widgets stack vertically on narrow viewports (NFR-MOB-001).
Role Permissions	All listed roles: Read.
Related Business Rules	BR-EXM-005, BR-EXM-007

Success/Error Condition	Message
Load Error	"Unable to load dashboard. Please retry."

SCR-EXM-002 — Exam Master

Purpose	Configure a new examination cycle: name, class, session, grading scheme, exam type, per FR-17.
User Roles	Academic Head, IT Admin
Navigation Path	Examination > Exam Master
Page Header	"Exam Master"
Breadcrumb	Home > Examination > Exam Master
Form Sections	Exam Configuration
Field Display Order	Exam Name -> Class -> Academic Session -> Grading Scheme -> Exam Date -> Exam Type
Conditional Visibility Rules	Grading Scheme options filtered to those matching the school's confirmed board affiliation.
UI Behaviour	Confirm Dialog: shown on Save if an identical exam name/class/session combination exists. Unsaved Changes Warning: yes.
Buttons & Actions	Save, Save & Configure Schedule, Cancel
Button Placement	Bottom-right of form.
Toolbar Actions	N/A
Table Columns	Exam Name, Class, Session, Status
Default Visible Columns	All columns visible by default
Hidden Columns	None
Row Actions	Edit, Configure Schedule, Delete (Configured status only)
Filters	Academic Session, Class, Status
Default Sorting	Exam Date, ascending
Default Pagination	25 rows/page
Bulk Actions	N/A
Export Formats	N/A (configuration screen)
Print Behaviour	N/A
Empty State	"No exams configured for this session."
Loading State	Loading indicator on save/load.
Popup/Dialog Details	Duplicate-exam confirm dialog.
Audit Information	Creation/edit logged with user and timestamp (Appendix-J Section 10).
Responsive Behaviour	Form fields stack single-column on narrow viewports.
Role Permissions	Academic Head: Create, Read, Update, Delete. IT Admin: full configuration authority.
Related Business Rules	BR-EXM-005 (schedule validity), BR-EXM-007 (grading scheme match)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Exam Name	Text	Yes	Non-empty	—	e.g., Term 2 Final	3/100	Editable
Class	Dropdown	Yes	Valid Class	—	Select class	—	Editable
Academic Session	Dropdown	Yes	Valid, non-closed session	Current session	—	—	Editable
Grading Scheme	Dropdown	Yes	Must match board affiliation (BR-EXM-007)	—	Select scheme	—	Editable
Exam Date	Date	Yes	Within Academic Session bounds	—	—	—	Editable
Exam Type	Dropdown	Yes	Unit Test/Term/Final/Other	—	—	—	Editable

Success/Error Condition	Message
Success	"Exam configuration saved."

Success/Error Condition	Message
Validation Error	"Please correct the highlighted fields."

SCR-EXM-003 — Exam Schedule

Purpose	Configure the subject-wise date/time schedule for a configured Exam.
User Roles	Academic Head, IT Admin
Navigation Path	Exam Master > Configure Schedule
Page Header	"Exam Schedule -- [Exam Name]"
Breadcrumb	Home > Examination > Exam Schedule
Form Sections	Subject Schedule List
Field Display Order	Subject -> Date -> Start Time -> End Time -> Max Marks
Conditional Visibility Rules	"Add Subject" row hidden once all Class subjects are scheduled.
UI Behaviour	Confirm Dialog: shown when removing a scheduled subject with marks already entered (blocked if any exist). Unsavd Changes Warning: yes.
Buttons & Actions	Add Subject, Remove, Save Schedule, Publish Schedule
Button Placement	"Add Subject" above list; "Save"/"Publish" bottom action bar.
Toolbar Actions	N/A
Table Columns	Subject, Date, Start Time, End Time, Max Marks
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit, Remove
Filters	N/A (single-exam context)
Default Sorting	Date, ascending
Default Pagination	Not paginated (typically <15 subjects)
Bulk Actions	N/A
Export Formats	PDF (printable schedule)
Print Behaviour	Print-friendly schedule layout for notice-board posting.
Empty State	"No subjects scheduled yet."
Loading State	Loading indicator on save.
Popup/Dialog Details	Date-clash warning dialog if two subjects overlap.
Audit Information	Schedule changes logged with user/timestamp.
Responsive Behaviour	Table scrolls horizontally on narrow viewports.
Role Permissions	Academic Head: Create, Read, Update, Delete, Publish. IT Admin: full authority.
Related Business Rules	BR-EXM-005 (schedule validity, no clashes)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Subject	Dropdown	Yes	Valid Subject for the Exam's Class	—	Select subject	—	Editable
Date	Date	Yes	Within Exam date bounds; no clash with another subject	—	—	—	Editable
Start Time	Time	Yes	Before End Time	—	—	—	Editable
End Time	Time	Yes	After Start Time	—	—	—	Editable
Max Marks	Number	Yes	Positive integer	100	—	—	Editable

Success/Error Condition	Message
Success	"Exam schedule saved and published."
Validation Error	"Subject date/time clashes with another subject."

SCR-EXM-004 — Subject Marks Entry

Purpose	Enter marks for all students in a class for one subject of an Exam, per FR-18.
---------	--

User Roles	Teacher, Academic Head (proxy entry)
Navigation Path	Examination > Subject Marks Entry
Page Header	"Marks Entry -- {Exam}, {Subject}"
Breadcrumb	Home > Examination > Marks Entry
Form Sections	Exam/Subject Selector, Student Marks List
Field Display Order	Exam -> Class/Section -> Subject -> Student Marks Grid
Conditional Visibility Rules	Row highlighted and flagged if entered marks deviate significantly from class average (anomaly flag, BR-EXM-006).
UI Behaviour	Auto Save: draft saved every 30 seconds. Confirm Dialog: shown on final Submit (locks marks per BR-EXM-003). Unsaved Changes Warning: yes.
Buttons & Actions	Save Draft, Submit for Verification, Cancel
Button Placement	Bottom action bar.
Toolbar Actions	Anomaly Filter Toggle (show flagged rows only)
Table Columns	Admission No, Student Name, Marks Obtained, Flag
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	N/A (inline edit)
Filters	N/A (single class/subject context)
Default Sorting	Roll order (Student Name, ascending)
Default Pagination	Not paginated (full class on one screen)
Bulk Actions	N/A
Export Formats	N/A (entry screen)
Print Behaviour	N/A
Empty State	"No students found for this class/section."
Loading State	Loading indicator while roster loads.
Popup/Dialog Details	Submit confirm dialog: "Submit marks for {n} students? This will lock entry pending verification."
Audit Information	Entered-by, timestamp per student logged (Appendix-J Section 10).
Responsive Behaviour	Grid becomes vertically stacked cards on narrow viewports.
Role Permissions	Teacher: Create (own assigned subject only). Academic Head: Create (proxy, any subject).
Related Business Rules	BR-EXM-002 (range validation), BR-EXM-003 (post-submit lock), BR-EXM-006 (anomaly flagging)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Exam	Dropdown	Yes	Valid, Active exam	—	Select exam	—	Editable
Subject	Dropdown	Yes	Valid Subject assigned to Teacher	—	Select subject	—	Editable
Marks Obtained (per student)	Number	No (Yes to submit)	0 <= value <= Max Marks (BR-EXM-002); blank = Absent	—	Enter marks	0/3 digits	Editable

Success/Error Condition	Message
Success	"Marks submitted for verification."
Validation Error	"Marks cannot exceed Max Marks." (BR-EXM-002)

SCR-EXM-005 — Bulk Marks Upload

Purpose	Upload marks for an entire class/subject via a single file instead of manual entry.
User Roles	Teacher, Academic Head, Admin Staff
Navigation Path	Subject Marks Entry > Bulk Upload
Page Header	"Bulk Marks Upload"
Breadcrumb	Home > Examination > Bulk Upload
Form Sections	Upload, Preview & Validate
Field Display Order	Exam -> Subject -> File Upload -> Preview Table
Conditional Visibility Rules	Preview table only appears after a file passes initial format validation.
UI Behaviour	Loading Indicator: shown during file parse and validation. Confirm Dialog: shown before committing parsed marks.

Buttons & Actions	Download Template, Upload File, Validate, Commit Marks, Cancel
Button Placement	"Download Template" top; "Commit"/"Cancel" bottom.
Toolbar Actions	N/A
Table Columns	Admission No, Student Name, Marks Obtained, Row Status (Valid/Error)
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit inline (correct a flagged row before commit)
Filters	Row Status (Valid/Error) to isolate problem rows
Default Sorting	Row Status, errors first
Default Pagination	Not paginated (full class preview)
Bulk Actions	Entire screen is a bulk-entry workflow.
Export Formats	CSV/Excel (template download)
Print Behaviour	N/A
Empty State	"No file uploaded yet."
Loading State	Progress indicator during parse/validation.
Popup/Dialog Details	Commit confirm dialog: "Commit marks for {n} valid rows? {m} rows have errors and will be skipped."
Audit Information	Upload event, uploaded-by, and per-row commit logged.
Responsive Behaviour	Preview table scrolls horizontally on narrow viewports.
Role Permissions	Teacher: Create (own subject). Academic Head/Admin Staff: Create (any subject, proxy).
Related Business Rules	BR-EXM-002 (range validation applies identically to bulk-uploaded marks)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Exam	Dropdown	Yes	Valid, Active exam	—	Select exam	—	Editable
Subject	Dropdown	Yes	Valid Subject	—	Select subject	—	Editable
File	File Upload	Yes	Supported format; row count matches class roster	—	Choose file	—	Editable

Success/Error Condition	Message
Success	"{n} marks committed successfully."
Validation Error	"{m} rows contain errors and were not committed."

SCR-EXM-006 — Marks Verification

Purpose	Review submitted marks, resolve anomaly flags, and lock a subject's marks, per BR-EXM-003.
User Roles	Academic Head
Navigation Path	Examination > Marks Verification
Page Header	"Marks Verification -- {Exam}, {Subject}"
Breadcrumb	Home > Examination > Verification
Form Sections	Submitted Marks, Anomaly Review
Field Display Order	Submitted Marks List -> Flagged Rows -> Lock Decision
Conditional Visibility Rules	"Lock Subject" disabled until every flagged row is either confirmed or corrected.
UI Behaviour	Confirm Dialog: mandatory before locking (irreversible without re-evaluation workflow, BR-EXM-003).
Buttons & Actions	Confirm Row, Correct Row, Lock Subject Marks
Button Placement	Inline per row; "Lock" bottom-right.
Toolbar Actions	Anomaly Filter Toggle
Table Columns	Admission No, Student Name, Marks, Flag, Status
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Confirm, Correct
Filters	Flag Status

Default Sorting	Flagged rows first
Default Pagination	Not paginated (full class)
Bulk Actions	Confirm All Non-Flagged Rows
Export Formats	PDF/Excel (verification summary)
Print Behaviour	N/A
Empty State	"No marks submitted yet for this subject."
Loading State	Loading indicator on load/lock.
Popup/Dialog Details	Lock confirm dialog: "Lock marks for {subject}? Further changes require a re-evaluation request."
Audit Information	Verification decision, verifier, timestamp logged per row and at subject-lock level.
Responsive Behaviour	Table condenses to essential columns on narrow viewports.
Role Permissions	Academic Head: Read, Update (confirm/correct), Lock. Teacher: no access.
Related Business Rules	BR-EXM-003 (lock authority), BR-EXM-006 (anomaly resolution)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Verification Remarks	Text	No -- Mandatory if correcting a flagged row	Non-empty when mandatory	—	Remarks	0/250	Editable

Success/Error Condition	Message
Success	"Subject marks locked."
Blocked	"All flagged rows must be resolved before locking."

SCR-EXM-007 — Result Processing

Purpose	Trigger grade/rank/GPA calculation for an Exam once all subjects are locked, per BR-EXM-001/BR-EXM-004.
User Roles	Academic Head
Navigation Path	Examination > Result Processing
Page Header	"Result Processing -- {Exam}"
Breadcrumb	Home > Examination > Result Processing
Form Sections	Subject Lock Status, Processing Trigger
Field Display Order	Subject Lock Status List -> Process Results button
Conditional Visibility Rules	"Process Results" disabled until every subject's lock status shows Locked.
UI Behaviour	Loading Indicator: shown during batch grade/rank calculation (background job for large classes). Confirm Dialog: shown before processing.
Buttons & Actions	Refresh Lock Status, Process Results
Button Placement	Bottom-right action bar.
Toolbar Actions	N/A
Table Columns	Subject, Lock Status, Marks Entered Count
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	N/A
Filters	N/A
Default Sorting	Subject, alphabetical
Default Pagination	Not paginated
Bulk Actions	N/A (processing is itself a batch operation)
Export Formats	N/A
Print Behaviour	N/A
Empty State	"No subjects configured for this exam."
Loading State	"Processing results..." with progress indicator.
Popup/Dialog Details	Confirm dialog: "Process results for {n} students? Grade, rank, and GPA will be calculated."

Audit Information	Processing run, initiator, timestamp, and outcome logged.
Responsive Behaviour	Table condenses on narrow viewports.
Role Permissions	Academic Head: Execute.
Related Business Rules	BR-EXM-001 (publish-gate precondition), BR-EXM-004 (GPA calculation)

Success/Error Condition	Message
Success	"Results processed for [n] students."
Blocked	"All subjects must be locked before processing." (BR-EXM-003)

SCR-EXM-008 — Grade Configuration

Purpose	Configure grade bands and GPA scale for a Grading Scheme, per BR-EXM-007.
User Roles	Academic Head, IT Admin
Navigation Path	Examination > Grade Configuration
Page Header	"Grade Configuration -- {Scheme Name}"
Breadcrumb	Home > Examination > Grade Configuration
Form Sections	Grading Scheme Details, Grade Band List
Field Display Order	Scheme Name -> Board Type -> Grade Band Rows (Grade, Min %, Max %, GPA Points)
Conditional Visibility Rules	"Save" disabled if any two grade bands overlap in percentage range.
UI Behaviour	Confirm Dialog: shown if editing a scheme already used by a closed exam (historical versioning applies). Unsaved Changes Warning: yes.
Buttons & Actions	Add Grade Band, Remove, Save Scheme
Button Placement	"Add" above list; "Save" bottom-right.
Toolbar Actions	N/A
Table Columns	Grade, Min %, Max %, GPA Points
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit, Remove
Filters	N/A
Default Sorting	Min %, descending
Default Pagination	Not paginated (typically <10 bands)
Bulk Actions	N/A
Export Formats	N/A (configuration screen)
Print Behaviour	N/A
Empty State	"No grade bands configured."
Loading State	Loading indicator on save.
Popup/Dialog Details	Overlap-validation error dialog.
Audit Information	Configuration changes logged per Appendix-F NFR-LOG-009.
Responsive Behaviour	Form stacks single-column on narrow viewports.
Role Permissions	Academic Head: Create, Read, Update. IT Admin: full authority, including historical-version management.
Related Business Rules	BR-EXM-007 (grading scheme integrity)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Scheme Name	Text	Yes	Non-empty, unique	—	e.g., CBSE Standard	3/100	Editable
Board Type	Dropdown	Yes	CBSE/ICSE/State Board	—	—	—	Editable
Grade	Text	Yes	Non-empty, unique within scheme	—	e.g., A1	1/5	Editable
Min %	Number	Yes	0-100; no overlap with adjacent bands	—	—	—	Editable
Max %	Number	Yes	0-100; > Min %	—	—	—	Editable
GPA Points	Number	Yes	Positive, per scale	—	—	—	Editable

Success/Error Condition	Message
Success	"Grading scheme saved."
Validation Error	"Grade bands must not overlap."

SCR-EXM-009 — Rank Calculation

Purpose	Review computed class rank and GPA output for an Exam before report cards are generated.
User Roles	Academic Head
Navigation Path	Result Processing > Rank Calculation
Page Header	"Rank Calculation -- {Exam}"
Breadcrumb	Home > Examination > Rank Calculation
Form Sections	Rank Result List
Field Display Order	Student Name -> Total Marks -> GPA -> Class Rank
Conditional Visibility Rules	Tied ranks displayed with a shared rank number and a tie indicator.
UI Behaviour	Read-only grid; no inline editing (recalculation only via Result Processing, SCR-EXM-007).
Buttons & Actions	Recalculate (returns to SCR-EXM-007), Export, Proceed to Report Cards
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Admission No, Student Name, Total Marks, GPA, Class Rank
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	View Student Result History (link to SCR-EXM-013)
Filters	Section
Default Sorting	Class Rank, ascending (default)
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel
Print Behaviour	Print-friendly rank list for internal review (not parent-facing).
Empty State	"Results have not been processed yet for this exam."
Loading State	Loading indicator on load.
Popup/Dialog Details	N/A
Audit Information	N/A (read view of Result Processing output).
Responsive Behaviour	Table scrolls horizontally on narrow viewports.
Role Permissions	Academic Head: Read, Export.
Related Business Rules	BR-EXM-004 (GPA/rank calculation)

Success/Error Condition	Message
Load Error	"Unable to load rank data."

SCR-EXM-010 — Report Card Generation

Purpose	Generate report card documents for students once results are processed, per FR-20.
User Roles	Academic Head, Admin Staff
Navigation Path	Examination > Report Card Generation
Page Header	"Report Card Generation -- {Exam}"
Breadcrumb	Home > Examination > Report Cards
Form Sections	Generation Scope, Generation Status
Field Display Order	Class/Section Selector -> Generate button -> Status List
Conditional Visibility Rules	"Generate" disabled until Rank Calculation (SCR-EXM-009) is complete for the selected scope.

UI Behaviour	Loading Indicator: shown during batch generation (background job for large classes).
Buttons & Actions	Generate Report Cards, Regenerate, Preview
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Admission No, Student Name, Generation Status, Generated Date
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Preview, Regenerate (individual)
Filters	Class, Section, Generation Status
Default Sorting	Student Name, ascending
Default Pagination	50 rows/page
Bulk Actions	Bulk Generate (entire class/section)
Export Formats	PDF (individual and batch)
Print Behaviour	Batch print-ready output for physical distribution.
Empty State	"No report cards generated yet."
Loading State	"Generating report cards..." with progress indicator.
Popup/Dialog Details	Confirm dialog: "Generate report cards for {n} students?"
Audit Information	Generation event, generated-by, timestamp logged (Document entity, Appendix-G/H).
Responsive Behaviour	Table condenses to essential columns on narrow viewports.
Role Permissions	Academic Head/Admin Staff: Create (generate), Read.
Related Business Rules	FR-20, BR-EXM-001 (publish-gate precondition)

Success/Error Condition	Message
Success	"Report cards generated for {n} students."
Blocked	"Rank calculation must be completed first."

SCR-EXM-011 — Publish Results

Purpose	Publish generated report cards to the Parent/Student portal, per BR-EXM-001.
User Roles	Academic Head
Navigation Path	Report Card Generation > Publish Results
Page Header	"Publish Results -- {Exam}"
Breadcrumb	Home > Examination > Publish Results
Form Sections	Publish Scope, Publish Confirmation
Field Display Order	Class/Section Selector -> Publish button
Conditional Visibility Rules	"Publish" disabled for any student whose report card has not yet been generated (SCR-EXM-010).
UI Behaviour	Confirm Dialog: mandatory before publish (triggers parent/student-facing visibility and notifications).
Buttons & Actions	Publish Selected, Publish All, Unpublish (Academic Head override only)
Button Placement	Bottom action bar.
Toolbar Actions	N/A
Table Columns	Admission No, Student Name, Report Card Status, Published (Y/N)
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Publish, Unpublish
Filters	Class, Section, Published Status
Default Sorting	Student Name, ascending
Default Pagination	50 rows/page
Bulk Actions	Bulk Publish (entire class/section)

Export Formats	N/A (publishing action, not an export)
Print Behaviour	N/A
Empty State	"No report cards ready to publish."
Loading State	Loading indicator during publish action.
Popup/Dialog Details	Confirm dialog: "Publish results for {n} students? Parents/students will be notified immediately."
Audit Information	Publish/unpublish event, actor, timestamp logged.
Responsive Behaviour	Table condenses on narrow viewports.
Role Permissions	Academic Head: Publish, Unpublish (override).
Related Business Rules	BR-EXM-001 (publish-gate enforcement)

Success/Error Condition	Message
Success	"Results published. Notifications sent."
Blocked	"Some selected students do not yet have a generated report card."

SCR-EXM-012 — Result Search

Purpose	Global quick search for a student's result by name, admission number, or exam.
User Roles	Academic Head, Teacher, Admin Staff, IT Admin
Navigation Path	Global Search Bar (available on all Examination screens)
Page Header	N/A (search widget)
Breadcrumb	N/A
Form Sections	N/A (search widget)
Field Display Order	Search Term -> Results Dropdown
Conditional Visibility Rules	Result set scoped to Teacher's assigned subjects/sections only for Teacher role.
UI Behaviour	Loading Indicator: shown while results load (debounced).
Buttons & Actions	Search, Clear
Button Placement	Inline with search field.
Toolbar Actions	N/A
Table Columns	N/A (compact dropdown: Student Name, Exam, Status)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	Open Student Result History
Filters	N/A
Default Sorting	Relevance (best match first)
Default Pagination	Top 10 matches; "View All" links to Student Result History
Bulk Actions	N/A
Export Formats	N/A
Print Behaviour	N/A
Empty State	"No matches found."
Loading State	Spinner within search dropdown.
Popup/Dialog Details	N/A
Audit Information	N/A (read-only search).
Responsive Behaviour	Search bar collapses to an icon on narrow viewports, expanding on tap.
Role Permissions	All listed roles: Read (scoped per role).
Related Business Rules	NFR-SRCH-001, NFR-PERF-004

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Search Term	Text	Yes (to trigger)	Minimum 2 characters	—	Search by name, admission no., or exam	2/100	Editable

Success/Error Condition	Message
No Matches	"No results found for this search term."

SCR-EXM-013 — Student Result History

Purpose	View a single student's results across all exams and academic sessions, per FR-21.
User Roles	Academic Head, Admin Staff, Teacher (own section), Parent/Student (own/linked, read-only)
Navigation Path	Student Profile > Results tab Examination > Student Result History
Page Header	"Result History -- {Student Name}"
Breadcrumb	Home > Examination > Result History
Form Sections	Exam-wise Summary, Subject-wise Detail
Field Display Order	Academic Session Selector -> Exam List -> Subject-wise Marks (expandable per exam)
Conditional Visibility Rules	Provisional-data label shown on any exam whose results are processed but not yet published (BR-RPT-004 pattern).
UI Behaviour	Expandable rows load subject detail on demand (lazy load).
Buttons & Actions	Download Report Card, Export History
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Exam Name, Session, GPA, Class Rank, Status
Default Visible Columns	All visible
Hidden Columns	Subject-wise marks (hidden until row expanded)
Row Actions	Expand/Collapse, Download Report Card
Filters	Academic Session
Default Sorting	Exam Date, descending (most recent first)
Default Pagination	20 rows/page
Bulk Actions	N/A
Export Formats	PDF (report card), Excel (full history)
Print Behaviour	Individual report card print-ready.
Empty State	"No exam results found for this student."
Loading State	Loading indicator on load/expand.
Popup/Dialog Details	N/A
Audit Information	N/A (read view; audit trail lives on underlying MarksRecord/ReportCard entities).
Responsive Behaviour	Exam list becomes a stacked card view on narrow viewports.
Role Permissions	Academic Head/Admin Staff/Teacher: Read (Teacher: own section). Parent/Student: Read (own/linked only, published results only).
Related Business Rules	BR-EXM-001 (published-only visibility for Parent/Student), FR-21

Success/Error Condition	Message
Load Error	"Unable to load result history."

SCR-EXM-014 — Result Analytics

Purpose	Trend and performance analytics across exams, classes, and subjects for Management and Academic Head.
User Roles	Academic Head, Management, IT Admin (read-only)
Navigation Path	Reports > Result Analytics
Page Header	"Result Analytics"
Breadcrumb	Home > Reports > Result Analytics
Form Sections	Analytics Filters, Trend Charts, Summary Table
Field Display Order	Report Scope -> Academic Session Range -> Class/Subject Filter -> Generate
Conditional Visibility Rules	Chart view replaces table view when Report Scope = Session-over-Session Trend.

UI Behaviour	Loading Indicator: shown during generation; large multi-session queries run as a background job (NFR-RPTP-001).
Buttons & Actions	Generate, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	Chart/Table View Toggle
Table Columns	Class/Subject, Average %, Pass %, Above-Threshold Count
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Student Result History)
Filters	Academic Session Range, Class, Subject
Default Sorting	Average %, descending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Chart and table export to print-ready PDF.
Empty State	"No data available for the selected filters."
Loading State	"Generating analytics..." with background-job notice for large ranges.
Popup/Dialog Details	N/A
Audit Information	Report generation logged (generated-by, timestamp, filters) per Appendix-J Section 15.
Responsive Behaviour	Charts resize to viewport; table becomes horizontally scrollable on narrow viewports.
Role Permissions	Academic Head/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002 (read-only reporting), BR-RPT-004 (provisional-data labelling), NFR-RPTP-001

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Scope	Dropdown	Yes	Class-wise/Subject-wise/Session-over-Session Trend	Class-wise	—	—	Editable

Success/Error Condition	Message
Success	"Analytics generated."
Queued	"Large analytics query queued -- you will be notified when ready."